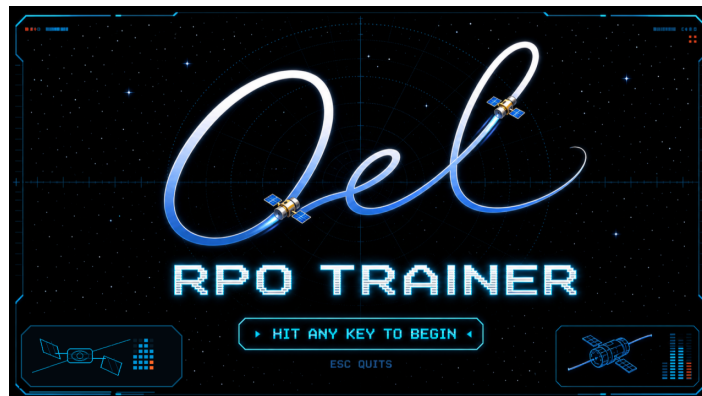


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Orbital Engagement Lab RPO Trainer

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Interactive RIC-frame intuition training for rendezvous and proximity operations education.

What It Is

The Orbital Engagement Lab RPO Trainer is a public, open-source educational tool that lets cadets manually command radial, in-track, and cross-track maneuvers, observe the resulting relative orbital motion, and review structured after-action debriefs tied to specific learning objectives.

It is not a replacement for classroom theory. It is a hands-on bridge between relative-motion equations, instructor explanation, and the physical intuition students need before RPO concepts feel natural.

Why This Helps

Students can solve relative-motion equations before they develop intuition for how spacecraft actually move near one another. Many first instincts come from aircraft, driving, or flat-Earth motion: thrust toward the target, stop when close, expect straight-line closure. Orbital relative motion does not reward those instincts.

The RPO Trainer gives cadets a safe, repeatable feedback loop:

- Predict what a maneuver should do;
- Command radial, in-track, or cross-track inputs;
- Observe the resulting RIC-frame motion;
- Compare the outcome against pass/fail criteria and debrief

compare the cadets against program criteria and metrics;

- Try again with better intuition.

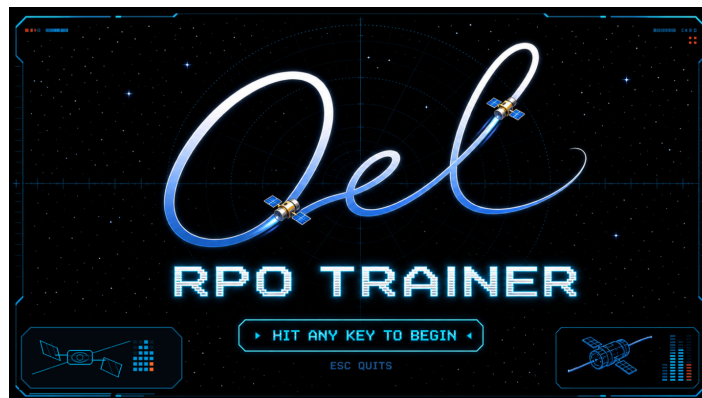
Because cadets directly control the spacecraft and receive immediate visual and metric feedback, the trainer can make abstract relative-motion behavior more memorable than static plots or worked examples alone.

Curriculum Fit

The trainer reinforces concepts commonly taught in astronautics and space operations courses:

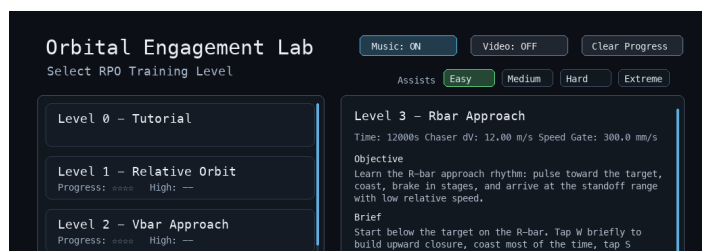
- Hill/RIC frame interpretation;
- natural relative motion;
- rendezvous geometry;
- V-bar and R-bar approach logic;
- delta-v management;
- passive safety and keepout awareness.

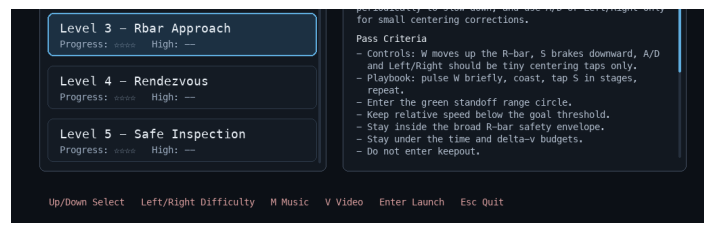
Visual Preview



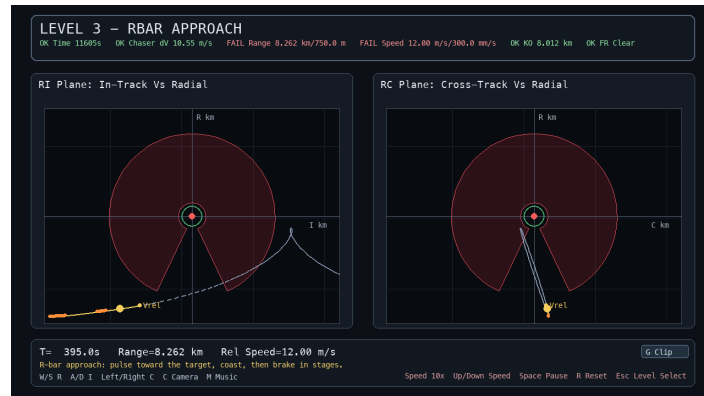
The trainer opens with a level selector and guided launch flow designed for local classroom use.

The level selector gives instructors and cadets a curated sequence of RPO training levels with objectives, pass criteria, assist difficulty, video recording, and progress controls.

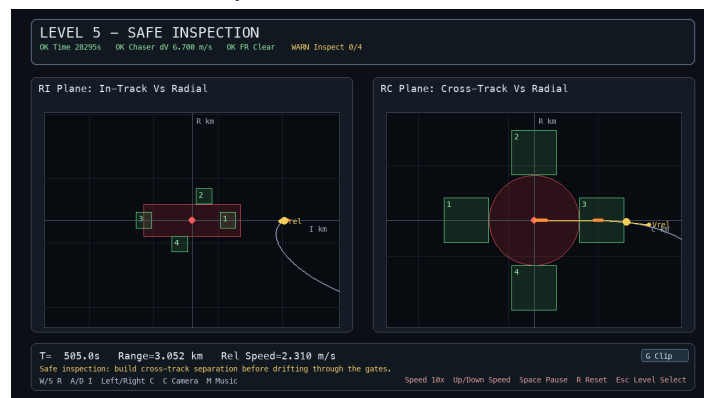




Level 3 emphasizes R-bar approach geometry, staged braking, keepout awareness, and the coupling between radial and in-track motion.



Level 5 asks students to build passively safer cross-track separation before drifting through inspection gates around a forbidden in-track cylinder.



Learning Objectives

Cadets can use the trainer to practice:

- RIC/Hill-frame relative-motion intuition.
- Natural motion, drift, and passive safety concepts.
- V-bar and R-bar approach discipline.
- Keep-out-zone and goal-region awareness.
- Terminal rendezvous with range and range-rate constraints.
- Delta-v budgeting and after-action review.

Suggested Classroom Use

A first classroom module can fit inside a short lab or lesson block:

Time	Activity
5 min	Instructor frames RIC axes, natural relative motion, and the training goal.
10 min	Cadets complete the guided tutorial.
15 min	Cadets attempt a focused level such as passive relative motion or V-bar approach.
10 min	Instructor leads a debrief using generated metrics, trajectory plots, and common mistakes.

The trainer can also support homework exploration, office-hour demos, or instructor-led live demonstrations during an astrodynamics or space operations lesson.

Included Training Modes

1. Guided tutorial for controls, RIC plots, pulse-and-coast behavior, and speed controls.
2. Ten structured RPO levels covering passive motion, V-bar/R-bar approaches, terminal rendezvous, eccentric-orbit cases, and defensive-target examples.
3. Sandbox mode for open-ended exploration.
4. Arcade pursuit mode for replayable practice.
5. Markdown/JSON debriefs and optional attempt recordings for structured training levels.

Example First Lab

Lab 1: Natural Relative Motion

Objective:

Explain why passive relative motion in orbit does not behave like straight-line translation in an inertial classroom sketch.

Student task:

Complete the guided tutorial, then attempt the passive relative-motion level.

Submit the generated debrief or selected screenshots showing the RIC trajectory, closest approach, final range, final relative speed, keepout result, and approximate delta-v.

Discussion prompts:

- Which parts of the motion matched your intuition?
- Which parts were surprising after you stopped thrusting?
- How did the RIC plots change your understanding of radial, in-track, and cross-track motion?
- What would you change on a second attempt?

Evidence Produced

Structured training attempts can produce debrief folders under:

Debriefs include:

- pass/fail result and miss reason;
- closest approach;
- final range;
- final relative speed;
- keep-out time or violation status;
- approximate delta-v;
- trajectory and timeline plots;
- control-command and cumulative delta-v plots;
- concise coaching observations.

These artifacts make the trainer useful for reflective learning: students can compare what they intended to do, what they commanded, and what the relative motion actually did.

Responsible Use

The RPO Trainer is an educational public beta. It is intended for intuition building, classroom demonstration, and exploratory learning with public, non-sensitive scenarios.

It is not flight-qualified software, an operational training system, a certification environment, mission-assurance evidence, or a substitute for formal astrodynamics instruction and validation.

Feedback Request

I would value instructor feedback on whether this could support RPO intuition-building in an astronautics or space operations classroom. Useful feedback would include:

- whether the first-lab flow matches real teaching needs;
- which levels best support existing course objectives;
- what terminology, diagrams, or debrief fields would make it more useful to cadets;
- what should be simplified before classroom use;
- what evidence an instructor would want students to submit.

Try It Locally

Install the game dependencies, then launch the level selector:

```
git clone https://github.com/adamcohen8/orbital-
engagement-lab.git
cd orbital-engagement-lab
python -m venv .venv
source .venv/bin/activate
.venv/bin/python -m pip install "[game]"
.venv/bin/python run_game.py
```

For project context, see docs/game-mode-roadmap.md.

Web Preview

A lightweight browser preview is available for quick demos without installing the Python package:

<https://adamcohen8.github.io/orbital-engagement-lab/>

The web preview includes the tutorial and sandbox-style RIC controls. The full local trainer includes the complete level set, scenario YAML support, recordings, and structured debrief reports.

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